

1.4 Working with Radicals - Worksheet

MCR3U

SOLUTIONS

1) Simplify

a) $3(4\sqrt{5})$

$$= 12\sqrt{5}$$

b) $\sqrt{5}(-2\sqrt{7})$

$$= -2\sqrt{35}$$

c) $2\sqrt{3}(3\sqrt{2})$

$$= 6\sqrt{6}$$

2) Express each as a mixed radical in simplest form

a) $\sqrt{12}$

$$= \sqrt{4} \times \sqrt{3}$$

$$= 2\sqrt{3}$$

b) $\sqrt{147}$

$$= \sqrt{49 \times 3}$$

$$= \sqrt{49} \times \sqrt{3}$$

$$= 7\sqrt{3}$$

c) $\sqrt{252}$

$$= \sqrt{36 \times 7}$$

$$= \sqrt{36} \times \sqrt{7}$$

$$= 6\sqrt{7}$$

3) Simplify

a) $2\sqrt{3} - 5\sqrt{3} + 4\sqrt{3}$

$$= 1\sqrt{3}$$

b) $11\sqrt{5} - 4\sqrt{5} - 5\sqrt{5} - 6\sqrt{5}$

$$= -4\sqrt{5}$$

c) $\sqrt{6} - 4\sqrt{2} + 3\sqrt{6} - \sqrt{2}$

$$= 4\sqrt{6} - 5\sqrt{2}$$

d) $2\sqrt{10} - \sqrt{10} - 4\sqrt{10} + \sqrt{5}$

$$= -3\sqrt{10} + \sqrt{5}$$

4) Add or subtract as indicated

a) $8\sqrt{2} - 4\sqrt{8} + \sqrt{32}$

$$= 8\sqrt{2} - 4(\sqrt{4}\sqrt{2}) + (\sqrt{16})(\sqrt{2})$$

$$= 8\sqrt{2} - 8\sqrt{2} + 4\sqrt{2}$$

$$= 4\sqrt{2}$$

b) $\sqrt{20} - 4\sqrt{12} - \sqrt{125} + 2\sqrt{3}$

$$= (\sqrt{4}\sqrt{5}) - 4(\sqrt{4}\sqrt{3}) - (\sqrt{25}\sqrt{5}) + 2\sqrt{3}$$

$$= 2\sqrt{5} - 8\sqrt{3} - 5\sqrt{5} + 2\sqrt{3}$$

$$= -3\sqrt{5} - 6\sqrt{3}$$

$$c) 5\sqrt{3} - \sqrt{72} + \sqrt{243} + \sqrt{8}$$

$$= 5\sqrt{3} - (\sqrt{36})(\sqrt{2}) + (\sqrt{81})(\sqrt{3}) + (\sqrt{4})(\sqrt{2})$$

$$= 5\sqrt{3} - 6\sqrt{2} + 9\sqrt{3} + 2\sqrt{2}$$

$$= 14\sqrt{3} - 4\sqrt{2}$$

$$d) \sqrt{44} + \sqrt{88} + \sqrt{99} + \sqrt{198}$$

$$= (\sqrt{4})(\sqrt{11}) + (\sqrt{4})(\sqrt{22}) + (\sqrt{9})(\sqrt{11}) + (\sqrt{9})(\sqrt{22})$$

$$= 2\sqrt{11} + 2\sqrt{22} + 3\sqrt{11} + 3\sqrt{22}$$

$$= 5\sqrt{11} + 5\sqrt{22}$$

5) Expand and simplify

$$a) 5\sqrt{6}(2\sqrt{3})$$

$$= 10\sqrt{18}$$

$$= 10(\sqrt{9})(\sqrt{2})$$

$$= 30\sqrt{2}$$

$$b) 8\sqrt{5}(\sqrt{10})$$

$$= 8\sqrt{50}$$

$$= 8(\sqrt{25})(\sqrt{2})$$

$$= 40\sqrt{2}$$

$$c) 11\sqrt{2}(5\sqrt{3})$$

$$= 55\sqrt{6}$$

6) Expand and simplify where possible

$$a) 3(8 - \sqrt{5})$$

$$= 24 - 3\sqrt{5}$$

$$b) \sqrt{3}(\sqrt{6} - \sqrt{3})$$

$$= \sqrt{18} - \sqrt{9}$$

$$= (\sqrt{9})(\sqrt{2}) - \sqrt{9}$$

$$= 3\sqrt{2} - 3$$

$$c) 8\sqrt{2}(2\sqrt{8} + 3\sqrt{12})$$

$$= 16\sqrt{16} + 24\sqrt{24}$$

$$= 64 + 24(\sqrt{4})(\sqrt{6})$$

$$= 64 + 48\sqrt{6}$$

7) Expand and simplify where possible

$$a) (\sqrt{2} + 5)(\sqrt{2} + 5)$$

$$= \sqrt{4} + 5\sqrt{2} + 5\sqrt{2} + 25$$

$$= 2 + 10\sqrt{2} + 25$$

$$= 27 + 10\sqrt{2}$$

$$b) (\sqrt{3} + 2\sqrt{2})(5 + 5\sqrt{2})$$

$$= 5\sqrt{3} + 5\sqrt{6} + 10\sqrt{2} + 10\sqrt{4}$$

$$= 5\sqrt{3} + 5\sqrt{6} + 10\sqrt{2} + 20$$

$$c) (1 + \sqrt{5})(1 - \sqrt{5}) \text{ D.O.S.}$$

$$= (1)^2 - (\sqrt{5})^2$$

$$= 1 - 5$$

$$= -4$$

$$d) (4 - 3\sqrt{7})(\sqrt{7} + 1)$$

$$= 4\sqrt{7} + 4 - 3\sqrt{49} - 3\sqrt{7}$$

$$= 1\sqrt{7} + 4 - 21$$

$$= \sqrt{7} - 17$$

8) Simplify

a) $\frac{1}{4}\sqrt{54} - \frac{1}{4}\sqrt{150}$

$$= \frac{1}{4}(\sqrt{9})(\sqrt{6}) - \frac{1}{4}(\sqrt{25})(\sqrt{6})$$

$$= \frac{3}{4}\sqrt{6} - \frac{5}{4}\sqrt{6}$$

$$= -\frac{1}{2}\sqrt{6}$$

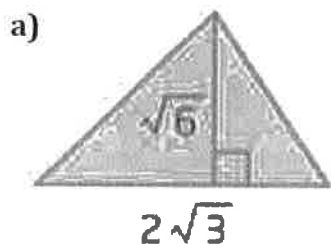
b) $\frac{1}{2}\sqrt{8} + \frac{3}{5}\sqrt{50} - \frac{2}{3}\sqrt{18}$

$$= \frac{1}{2}(\sqrt{4})(\sqrt{2}) + \frac{3}{5}(\sqrt{25})(\sqrt{2}) - \frac{2}{3}(\sqrt{9})(\sqrt{2})$$

$$= 1\sqrt{2} + 3\sqrt{2} - 2\sqrt{2}$$

$$= 2\sqrt{2}$$

9) Find a simplified expression for the area of each shape



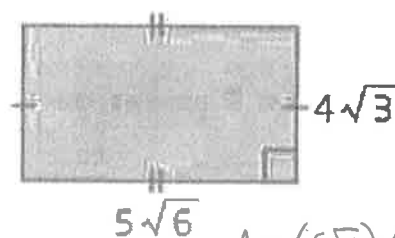
$$A = \frac{\sqrt{6}(2\sqrt{3})}{2}$$

$$= \sqrt{18}$$

$$= (\sqrt{9})(\sqrt{2})$$

$$= 3\sqrt{2}$$

b)



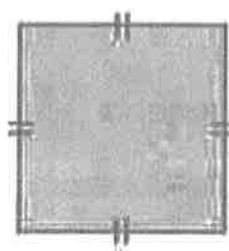
$$A = (5\sqrt{6})(4\sqrt{3})$$

$$= 20\sqrt{18}$$

$$= 20(\sqrt{9})(\sqrt{2})$$

$$= 60\sqrt{2}$$

c)



$$A = (3\sqrt{5})^2$$

$$= 9(5)$$

$$= 45$$

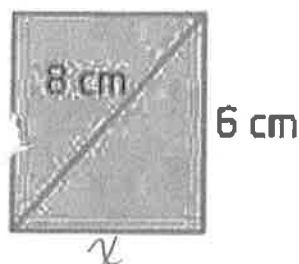
d)



$$A = \pi(\sqrt{2})^2$$

$$= 2\pi$$

10) Find the area and perimeter of the rectangle shown. Express your answer in simplified radical form.



$$x^2 + 6^2 = 8^2$$

$$x^2 = 64 - 36$$

$$x = \sqrt{28}$$

$$x = 2\sqrt{7}$$

$$A = (2\sqrt{7})(6)$$

$$= 12\sqrt{7} \text{ cm}^2$$

$$P = 2(2\sqrt{7}) + 2(6)$$

$$= 4\sqrt{7} + 12 \text{ cm}$$

11) Simplify each of the following

a) $\frac{21-7\sqrt{6}}{7}$

$$= \frac{7(3-\sqrt{6})}{7}$$

$$= 3 - \sqrt{6}$$

b) $\frac{12-\sqrt{48}}{4}$

$$= \frac{12 - (\sqrt{6})(\sqrt{8})}{4}$$

$$= \frac{12 - 4\sqrt{3}}{4}$$

$$= \frac{4(3-\sqrt{3})}{4}$$

$$= 3 - \sqrt{3}$$

Answers

1) a) $12\sqrt{5}$ b) $-2\sqrt{35}$ c) $6\sqrt{6}$

2) a) $2\sqrt{3}$ b) $7\sqrt{3}$ c) $6\sqrt{7}$

3) a) $\sqrt{3}$ b) $-4\sqrt{5}$ c) $4\sqrt{6} - 5\sqrt{2}$ d) $-3\sqrt{10} + \sqrt{5}$

4) a) $4\sqrt{2}$ b) $-3\sqrt{5} - 6\sqrt{3}$ c) $14\sqrt{3} - 4\sqrt{2}$ d) $5\sqrt{11} + 5\sqrt{22}$

5) a) $30\sqrt{2}$ b) $40\sqrt{2}$ c) $55\sqrt{6}$

6) a) $24 - 3\sqrt{5}$ b) $3\sqrt{2} - 3$ c) $64 + 48\sqrt{6}$

7) a) $27 + 10\sqrt{2}$ b) $5\sqrt{3} + 5\sqrt{6} + 10\sqrt{2} + 20$ c) -4 d) $-17 + \sqrt{7}$

8) a) $-\frac{1}{2}\sqrt{6}$ b) $2\sqrt{2}$

9) a) $3\sqrt{2}$ b) $60\sqrt{2}$ c) 45 d) 2π

10) area = $12\sqrt{7}$ cm²; perimeter = $12 + 4\sqrt{7}$ cm

11) a) $3 - \sqrt{6}$ b) $3 - \sqrt{3}$